



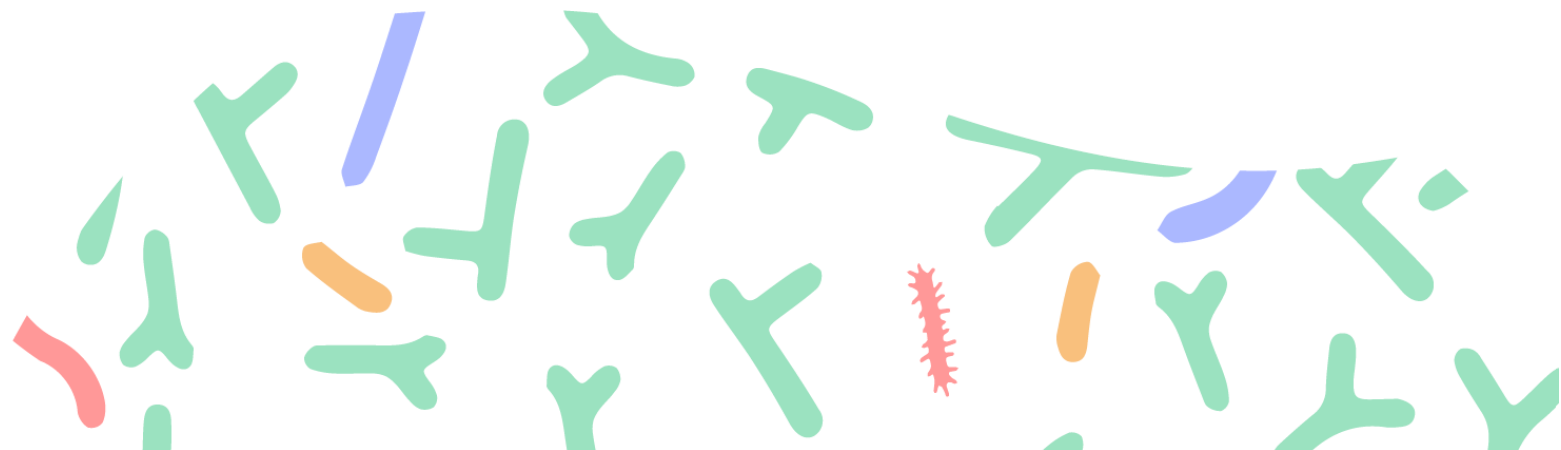
MICROBIOME HEALTH RESULTS REPORT

Physician Name:

Patient Name:

Kit ID:

Sampling Date:



RESULTS & INTERPRETATION

This report summary is provided by FRX to assist healthcare practitioners to interpret the microbiome health results report. The report should be used only after the health practitioner has conducted a full patient assessment which should include existing medications, allergies & intolerances and full patient medical history.

The main sections of this report are:

Microbiome diversity:	Provides the Shannon diversity which measures the overall community found in the gut, and the abundance of host DNA identified in the sample as important health metrics to monitor. This section also includes the enterotype which categorizes the microbial community based on its more dominant members.
Key Insights:	Describes important taxonomical and functional metrics associated with overall microbiome health and dysbiosis.
Microbiome Breakdown:	Provides the Firmicutes/Bacteroidetes & <i>Prevotella</i> / <i>Bacteroides</i> ratios as important composition metrics to monitor. This section also includes the list of microorganisms detected in the sample, their abundance and a description of their role.
Action Plan:	Provides interventional recommendations such as supplements, diet and lifestyle changes based on the results of the microbiome health test. This section also highlights some important medical and self-reported outcomes which may have a major impact on the gut microbiome.

SUMMARY

DIVERSITY & HOST DNA

page 3

Shannon Index

Host DNA content

KEY INSIGHTS

page 4-8



Great Results



Average Results



Needs Support

MICROBIOME BREAKDOWN

page 9-10

● Firmicutes

● Prevotella

● Bacteroidetes

● Bacteroides

F/B Ratio

P/B Ratio

● Beneficial

● Variable

● Unfriendly

● Unknown

MICROBIOME DIVERSITY

Shannon Diversity & Index

The Shannon diversity is a measure of amounts and types of microbial species in the gut. The more variety of species present, the greater likelihood of a healthy metabolism and robust immune system.

This diversity can be calculated using the Shannon index to determine the richness (variety of different species) and evenness (amount distribution of species) of the microbiome.

✓ **The higher the overall diversity/index the better!**

Shannon diversity

7

12

Host DNA content

Faecal samples collected for gut microbiome analysis are made-up of a biomass composed of microorganisms, undigested plant-based matter and a usually small amount of host cells that were passed during transit. Depending on the health status of the individual, surgical/pharmaceutical interventions and/or infections, the structural integrity of the intestinal lining may be compromised and shed an unusually high amount of cells in the faecal sample. This can be measured by the abundance of human (host) DNA identified when sequencing the total DNA in the sample.

✓ **The lower the host DNA content the better!**

Host DNA content

>5%

Enterotype

The Enterotype is determined based on the dominating microbiota members' overall impact on the gut as part of the community. This may change over time, in particular after experiencing major changes in diet, lifestyle, environment, medication or surgery. Maintaining the same Enterotype over time is a sign of microbiome stability, which is healthy assuming the diversity stays high, and the types of species present are mostly beneficial.

Enterotype

KEY INSIGHTS



Microbial abundance (pp. 4-7)

Insights based on the relative abundance of certain beneficial and pathogenic microbes.

Functional potential (pp. 7-8)

Insights based on the abundance of certain microbial genes representing the microbiome's genetic potential to perform a particular function.

Other metrics (pp. 8)

Insights calculated from a community of species or functional measures.

KEY INSIGHTS *(continued)*



(f) family; (g) genus; (s) species

MICROBIOME BREAKDOWN

Firmicutes/Bacteroidetes and *Prevotella*/*Bacteroides* ratios

Firmicutes & Bacteroidetes are the most abundant phyla of the human gut microbiome. Many studies have previously associated differences in this ratio with general gut dysbiosis, which may in turn be implicated in health status and a number of non-communicable disease.

Prevotella & *Bacteroides* are common genera found in the human gut microbiome and associated with plant-based vs Western industrialized diets, respectively. Differences in this ratio (higher P/B ratio) have been associated with weight-loss success.

Although neither ratios are considered measures of health or disease, they can be beneficial markers to monitor over time.

Firmicutes/Bacteroidetes (F/B) ratio

- Firmicutes
- Bacteroidetes

F/B ratio

Prevotella/*Bacteroides* (P/B) ratio

- Prevotella*
- Bacteroides*

P/B ratio

Microbiome Profile

Beneficial Variable Unfriendly Unknown

MICROBIOME BREAKDOWN *(continued)*

Top 20 most abundant species

Species	Abundance	Highlights
---------	-----------	------------